

FedDGHD: Diffusion Model-based Data Generation for Non-IID Challenges in Federated Learning

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Abstract—Federated Learning (FL) allows participants to collaboratively train a global model without sharing local data. However, the non-independent and identically distributed (non-IID) nature of data in real-world scenarios leads to a decline in the accuracy of existing federated learning methods. Using Generative Adversarial Networks (GANs) to generate realistic pseudo-data is a feasible approach to alleviate the non-IID issue. However, the realism of GAN-generated data comes at the cost of diversity and is prone to falling into local minima. Therefore, this paper innovatively integrates the pre-trained stable diffusion model's text-to-image mechanism with federated learning, constructing a client-server collaborative text generation mechanism that effectively alleviates the issue of data distribution bias. Furthermore, the effectiveness of FedDGHD is validated on datasets such as CIFAR10 and CIFAR100.

Index Terms—Federated Learning, Heterogeneous Data Large-scale Model

I. Introduction

Federated Learning [1] is a distributed learning technology that allows multiple participants to collaboratively train a model without sharing raw data while protecting data privacy. Among them, FedAvg (Federated Averaging Algorithm) [1] is a commonly used algorithm in federated learning, which forms a global model by averaging all the local models trained by the clients. However, due to the diverse and varied sources of client data, the data exhibits non-independent and identically distributed (Non-IID) characteristics. This can cause the local model weights to diverge in different directions, thereby affecting the generalization ability of the global model, as confirmed in Li's research [2]. Therefore, one of the biggest challenges in federated learning is how to obtain a model with good generalization ability under the Non-IID condition. Heterogeneous Federated Learning is a distributed machine learning method that takes into account the heterogeneity of client data and devices, aiming to address the challenges

of decreased model generalization ability caused by Non-IID distribution and device heterogeneity. This is also the focus of this work: effectively addressing the issue of model degradation caused by the distribution of local model weights.

Recently, researchers have proposed various solutions to address the issue of Non-IID in federated learning, mainly focusing on data augmentation. Initially, researchers attempted to make client sample features closer to independent and identically distributed by constructing shared datasets. However, this approach might increase training bias and privacy risks due to distribution differences and quality issues between shared and local datasets. Later, researchers tried to introduce Generative Adversarial Networks (GANs) into FL to generate high-quality synthetic data for dataset augmentation. This helps improve model generalization capabilities in cases of uneven data distribution or data sparsity while protecting privacy. However, GANs have higher complexity, require more computational resources, and increase the difficulty of training. Moreover, due to the non-convex optimization problem during the training process, they tend to get stuck in local minima, resulting in poor diversity of generated samples and limiting their potential applications. Although GANs can generate high-quality samples, this is at the expense of diversity.

Recently diffusion models (DM) [6] [18] have demonstrated superior performance in generating data, providing an opportunity for us to evolve. The famous Diffusion model is Stable Diffusion model. The Stable Diffusion model is a latent-based diffusion model proposed by Rombach [6] in 2022, which can realize the Vincenless graph and graph-generating tasks. The pre-trained DM model has strong prior knowledge and can generate data with sufficient diversity in categories and distributions. However, the ability of the pre-trained Stable Diffusion

model to solve heterogeneous federated learning has not been explored.

Based on the above, we propose a new data augmentation framework—FedDGHD. This framework is a method based on a pre-trained Stable Diffusion model. It introduces a pre-trained SD model [6] to generate abstract descriptions after understanding image data through image-to-text processing. These descriptions form a description set, which is then uploaded to the server. The server uses this description set with a diffusion model to generate a synthetic dataset. During the aggregation phase, the global model is fine-tuned using the generated dataset through distillation and contrastive techniques.

Our contributions are as follows:

- In the innovation of data heterogeneity in federated learning, we introduce a pre-trained SD model and text-to-image models to generate realistic and diverse images, balancing the data distribution across clients. Compared to methods that use GANs, our approach produces more realistic and diverse generated data. To the best of our knowledge, we are the first to explore the use of a pre-trained SD model’s powerful image generation capability, combined with the text-to-image generation-Diffusion model, to address the challenges of heterogeneous federated learning.
- We propose a framework called FedDGHD, which utilizes the SD model to generate rich data to address the Non-IID distribution. At the same time, we design a feature-layer contrastive method during the aggregation phase to enable the global model to learn the rich knowledge from the DM, improving the model’s ability to distinguish between positive and negative samples and enhancing its generalization capability.
- We verify FedDGHD’s effectiveness on CIFAR10, CIFAR100, and other datasets. At the same time, visualization experiments show that our method can generate high-quality and diverse datasets with the same quality as the local datasets and does not leak sensitive information.

II. Related work

A. Heterogeneous Federated Learning

HFL (Heterogeneous Federated Learning) aims to address the challenges brought by heterogeneity in federated learning, including but not limited to the non-iid (non-independent and identically distributed) problem and the differences between edge devices in the system, which hinder the coordinated training in federated learning. In the research on the non-iid distribution problem, most of the work focuses on expanding the data sample. These methods usually set up a global shared knowledge set or generate pseudo data to enrich the features of the client’s data. For example, FRAug [3] trains a shared generator to enhance the training space of each client. Prashant [4] et al. enhance the local data set’s images in different ways

to enrich the features of the client’s data. He [5] and Liu [9] share part of the local data to let the model have more knowledge to learn. FedFTG [14], DFRD [15], and FedGen [16] train an auxiliary generator using the idea of generative adversarial networks to generate diverse and highly similar samples to the local data set. However, these methods have problems. First, the public data set may have a bias and have varying quality; second, training with adversarial generation models is difficult, and the non-convex optimization problem in the training process may limit the sample diversity, limiting its potential application in fields such as image generation.

B. Diffusion Models

Recently, diffusion models (DMs) have achieved significant success in the field of image generation, offering a novel approach to image synthesis. The Stable Diffusion model, proposed by Rombach et al [17]. in 2022, is based on the diffusion process in latent space and effectively addresses issues of data distribution bias, resulting in diverse and high-quality images. This model not only accomplishes tasks such as text-to-image and image-to-image generation but also utilizes conditions like text to guide the creation of specific images. Pre-trained diffusion models can generate a variety of images that conform to data distribution within a reasonable time and computational cost. Dhariwal points out that since GANs rely on adversarial training, this can lead to model convergence failure and vanishing gradients, especially when the generator produces plausible outputs, which can easily result in getting stuck in local minima. Moreover, if the discriminator fails to find the best strategy, the generator’s iterative process will tend to provide outputs that are reasonable for the current discriminator, limiting the model’s potential for innovation. Therefore, diffusion models, with their outstanding performance and innovative potential in image generation, are gradually becoming an important research direction in the field of computer vision. Consequently, Stable Diffusion has broad application prospects in the field of FL, especially in HFL. We propose to apply Stable Diffusion to FL to address the challenges posed by heterogeneous data.

III. SCHEME DESIGN

In this section, we will delve into our proposed FedDGHD method. First, we will provide an overview of the system model, followed by a detailed description of the specific processes and objectives of each step.

A. General Framework

We have introduced a method called FedDGHD, based on the pre-trained Stable Diffusion (SD) model, designed to enhance the generalization capability of models. FedDGHD leverages the SD model to augment training data, thereby reducing the heterogeneity of non-identically and independently distributed (Non-IID) data, and consequently improving the model’s generalization. This

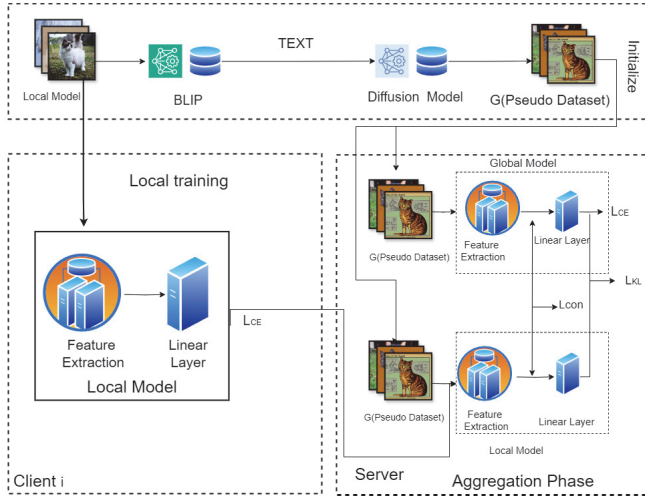


Fig. 1. General framework of FedDGHD

method is divided into three parts: data generation, local training, and global aggregation. In the data generation phase, clients upload text-described datasets to the server and the server uses the SD model to generate a diverse augmented dataset. In the local training phase, the clients train their models and upload the parameters to the server. For the global aggregation phase, we employ knowledge distillation and contrastive learning methods to fine-tune the global model, enabling it to more effectively absorb the knowledge from each client's local dataset. Additionally, by utilizing the SD model's knowledge to expand the samples, we mitigate the impact of Non-IID on the global model's generalization ability, thereby enhancing the model's overall performance. The overall framework diagram of FedDGHD is shown in Fig.1.

B. Client-server Collaborative Text Generation Mechanism

In the data generation step, the client first uses the BLIP2 model to generate text for images in the local dataset. The BLIP2 image-text pre-training model freezes the image encoders of pre-trained image models and language models, while introducing a trainable Q-Former mechanism between them to effectively align different modalities and enhance the model's understanding of the relationship between images and text. Compared to other multimodal models, BLIP2 stands out in tasks such as visual question answering and image description generation. When generating image descriptions, BLIP2 focuses on extracting the core visual features of the image and generating simple text descriptions. This method ensures the accuracy and relevance of the descriptions without leaking sensitive information. Therefore, we use BLIP2 for text description, and then transmit this text to the SD model to generate synthetic data, ensuring the quality of the generated images. With the BLIP-DIFFUSION combined model, controllable images can be

generated without disclosing sensitive information. This process is shown in lines 2 to 10 of Algorithm 1.

C. Model Aggregation

In contrastive learning, both the global model and the local model project features into the contrastive subspace through the projection layer. By leveraging the powerful prior knowledge of the stable diffusion model, we generate data that exhibits both diversity and high fidelity. These samples include not only those that are easily distinguishable by category but also those that are difficult to differentiate. Next, we construct anchor-positive-negative sample pairs using the generated dataset. The global and local embeddings of same-class, same-origin samples serve as positive pairs, while the global and local embeddings of different-class samples serve as negative pairs. We build an intra-class and inter-class dual-path contrastive loss function for the local-global model, optimizing both global knowledge alignment and the sharpening of local boundaries in the feature space. The intra-class loss maximizes the similarity between positive sample features, while the inter-class loss enhances the distinguishability of heterogeneous samples through adaptive boundary constraints. This method allows the global model to learn from the powerful prior knowledge of the diffusion model during fine-tuning, while also improving its ability to distinguish between intra-class and inter-class features. The specific contrastive formulas are given by equations (1), (2), (3), and (4), where (x_p^+) represents the positive sample related to x_p , and x_p^- represents the negative sample related to x_p . τ is the temperature coefficient, G is the generated dataset, M is the number of clients, and P is the projection layer of the model.

At the same time, using knowledge distillation techniques, the global model is treated as the student model and the local models as teacher models. During fine-tuning, the global model learns from the soft and hard labels of the local models, quickly integrating the knowledge from all local models. Additionally, by using model distillation, the aggregation speed can also be accelerated. This dual approach allows the model to prioritize local knowledge while leveraging generated knowledge to balance data features and optimize model performance. The distillation loss is given by Equation (5), where $W^g(x_p)$ is the global model's prediction distribution for sample x_p , and $W^i(x_p)$ is the local model's prediction distribution for sample x_p .

$$A = \exp \{ (P(W_g(x_p)) \cdot P(W_i(x_p))) / \tau \} \quad (1)$$

$$B = \exp \{ P(W_i(x_p^-)) \cdot P(W_g(x_p)) / \tau \} \quad (2)$$

$$C = \exp \{P(Wi(x_p^+)) \cdot P(Wg(x_p^+)/\tau)\} \quad (3)$$

$$L_3 = \sum_{i=1}^M \sum_{(x_p, y_p) \in G} \log \left(\frac{A + C}{A + C + B} \right) \quad (4)$$

$$L_2(W^i(x_p) || W^g(x_p)) = \sum_{(x_p, y_p)}^G \sum_{i=1}^M W^i(x_p) \log \left(W^i(x_p) / W^g(x_p) \right) \quad (5)$$

Finally, the Cross-Entropy Loss of the global model on the pseudo-data is computed.

The training process is detailed in lines 11 to 27 of Algorithm 1.

IV. Experiments

A. Experiments Setting

We conduct performance evaluation for federated learning on CIFAR-10 and CIFAR-100 datasets. CIFAR-10 consists of 10 categories, each containing 6,000 images, with each image sized at 32×32 pixels. CIFAR-100 contains 100 categories, each with 600 images. We compare our proposed federated learning algorithm with five popular federated learning algorithms, namely FedAvg, Moon [8], FedGen, FedProx, and FedPCL. To ensure the consistency of the experiments, we set the global training rounds uniformly to 200 rounds. The number of clients is set to 20 and each client performs 1 round of local training with batch B=64, learning rate $lr=\{0.1, 0.001\}$, $\alpha=\{1, 0.5, 0.3\}$, and Stochastic Gradient Descent (SGD) optimizer is used for local training. We use two different models in our experiments: one contains two 5×5 convolutional layers and two fully connected layers, referred to as CNN1; the other comprises four convolutional layers (two 5×5 convolutional layers and two 3×3 convolutional layers) followed by two fully connected layers, referred to as CNN2. The experiments are conducted on a server equipped with an NVIDIA® V100 Tensor Core GPU.

B. Experiment Results

Performance Comparison: The data sets CIFAR10 and CIFAR100 are used to compare the performance of FedDGHD and other methods. In both training sets, the hyperparameters for the servers to train in the FedDGHD aggregation phase are set as follows. The aggregate learning rate $lr_g = 0.001$, the aggregate training batch $B_g = 8$, the contrast temperature coefficient $\tau = 0.13$, the contrast loss ratio $\eta = 0.0001$, the distillation loss ratio $\gamma = 0.15$, and the aggregate learning rate $lr_g = 0.001$. The cross-entropy coefficient $\beta = 1$, where $\eta = 0.1$ and $\tau = 2$ are used under the CNN2 model. The first to seventh rows of Table I compare the performance of FedDGHD with the baseline under different α distributions when using

Algorithm 1 Training procedure of FedDGHD

1: Input:

- M : Number of clients
- K : Number of classes/categories
- N : Number of samples/instances in dataset
- W^i : Local models for each client $i \in M$
- D^i : Data distributors for each client i
- μ_1 : Local training learning rate
- μ_2 : Global training learning rate
- G : Pseudo dataset (x_p, y_p)
- T_1 : Global training epochs
- T_2 : Local training epochs
- W^B : Blip model
- W_t^g : Global model when epoch=t
- W^{DM} : Diffusion model

2: Initialization:

3: Client: Initialize local models

4: for each client $i \in M$ in parallel do

5: $D_i^t = W^B(D^i)$

6: end for

7: Server: Generate pseudo dataset

8: for each client $i \in M$ in parallel do

9: $G = W^{DM}(D_i^t)$

10: end for

11: Training:

12: Client: Perform local training

13: for each client $i \in M$ in parallel do

14: for $n = 1$ to T_2 do

15: $W_n^i = W_{n-1}^i + \mu_1 \nabla L_{CE}(W_{n-1}^i, D^i)$

16: end for

17: return W_n^i

18: end for

19: Server:

20: for $t = 1$ to T_1 do

21: Compute L_1 : Cross-entropy loss

22: Compute L_2 : Distillation loss

23: Compute L_3 : Cross-Entropy Loss

24: $L = \beta L_1 + \gamma L_2 + \eta L_3$

25: $W_t^g = W_{t-1}^g + \mu_2 \nabla L(W_{t-1}^g, G)$

26: end for

27: return W_t^g

CIFAR100 and CIFAR10. In the table, the local learning rate is $lr=0.001$ and $B=64$ for the 2nd and 3rd rows, while for the 4th and 5th rows, the local learning rate is $lr=0.01$ and $B=64$. According to the results in the table, FedDGHD outperforms other methods in the vast majority of scenarios.

To validate the applicability of the FedDGHD method under different federated scales, we conducted multiple comparative experiments on the CIFAR10 dataset ($\alpha=0.5$). As shown in Fig. 2, when the number of clients gradually increases from 20 to 50, the accuracy fluctuation of FedDGHD is minimal, indicating that the method can

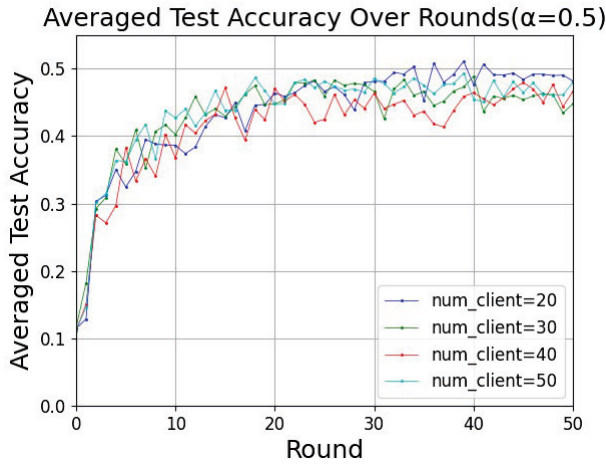


Fig. 2. Accuracy comparison of edDGHD with different number of clients

effectively adapt to typical node dynamic changes in federated learning systems and maintain reliable performance even in large-scale distributed deployments.

C. Discussion

Image Generation Privacy Issues: The use of textual descriptions in federated learning may pose privacy risks, so we specifically evaluate whether the generated images from text contain any privacy-sensitive information. As shown in Fig. 3(b), through comparisons based on five privacy-related categories from CIFAR-100 (such as "boy" and "baby"), the generated images do not share exact features with the original client images. Instead, they merely reflect general attributes without revealing any sensitive or personally identifiable information from the original data.

Text Extraction Privacy Issues: In this approach, clients are required to send their local text datasets to the server, which could potentially increase the risk of local data privacy exposure. However, the BLIP2 used by FedDGHD focuses on extracting the core visual features of images without paying attention to any sensitive information that might be implied within the images. This ensures the accuracy and relevance of the descriptions without leaking sensitive information. In Fig. 3(a), we present the images from the local dataset along with the text descriptions extracted by the BLIP model. It can be observed that the text extracted by BLIP only provides a simple categorical description of the main subject in the image and offers a brief explanation of the subject's position or background within the image, without delving into detailed descriptions of the subject's features. Therefore, FedDGHD does not disclose sensitive information.

Additional observations: In the aggregation stage, we use the pseudo-data set to compare the feature projections of the local and global models, so as to use the powerful knowledge of the SD model to improve the global model's

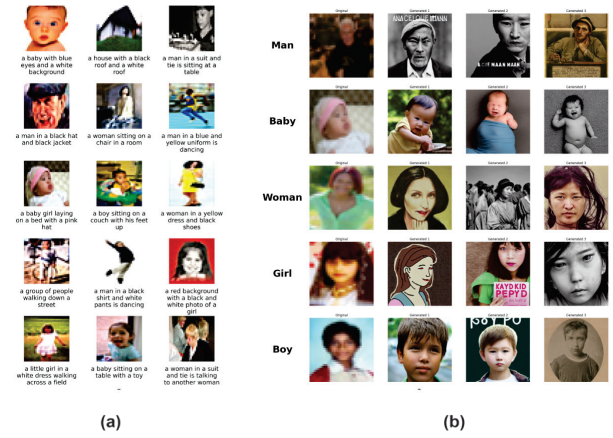


Fig. 3. Visualization of generated images in FedDGHD. (a) Visualization of the image-to-text step, showing images and generated text. (b) Comparison between original and generated images, indicating that the generated images are only similar in style to the original client images, thus not disclosing any privacy in the original images.

understanding of the differences between positive and negative samples, thereby enhancing its generalization ability. To test this, we compare the generalization performance of FedDGHD with and without contrastive loss. The experimental results, as shown in the seventh column of Table I, indicate that the contrastive method effectively improves the model's performance.

Resource Requirements: We compared the memory required for the runtime and the number of tensors the model needs to compute for each method, as shown in the eighth and ninth rows of Table I. The memory footprint of FedDGHD is similar to that of FedAvg, indicating that FedDGHD has a relatively low operational burden, with modest hardware requirements, making it easier to deploy. Meanwhile, the number of model tensors reflects the complexity of the model and the size of the data, which to some extent reflects the computational resource requirements and serves as an indicator for evaluating computational resources. From Table I, it can be seen that the complexity and data volume of FedDGHD are relatively small, which suggests that its computational resource requirements are not very high.

V. Conclusion

This paper proposes an innovative federated learning framework, which combines the text-guided image generation ability with the strong representation prior of the Stable Diffusion (SD) model, and combines feature comparison and knowledge distillation to improve the generalization of the model, so as to cope with the challenge of Non-independent and identical Distribution (Non-IID) of client data in federated learning. The future research direction will focus on two aspects: one is to design a lightweight SD model distillation strategy to reduce the computation and communication overhead, and the other is to explore the privacy-utility balance between

TABLE I
Performance results of FedDGHD compared with other methods on cifar10 and cifar100 datasets

Method	FedAvg	Fedprox	Moon	Fedgen	Fedpcl	FedDGHD	FedDGHD (No contrast)
Cifar10(CNN1) $\alpha=1$	0.3426	0.3412	0.3440	0.3429	0.3086	0.4248	*
Cifar10(CNN1) $\alpha=0.5$	0.3224	0.3220	0.3218	0.3488	0.3313	0.3823	0.3498
Cifar10(CNN1) $\alpha=0.3$	0.2885	0.2945	0.2949	0.2946	0.3029	0.3113	0.2766
Cifar10(CNN2) $\alpha=0.5$	0.5232	0.5362	0.5359	0.5369	0.4917	0.5457	*
Cifar10(CNN2) $\alpha=0.3$	0.4481	0.4517	0.4570	0.4662	0.4565	0.4745	*
Cifar100(CNN1) $\alpha=0.5$	0.2814	0.2913	0.2833	0.2879	0.0439	0.2979	0.2815
Cifar100(CNN2) $\alpha=0.3$	0.2689	0.2713	0.2739	0.2714	0.0631	0.2841	0.2644
Memory required to run(M)	937.25	1382.4	1402.88	938.34	937.78	962.58	*
Number of tensors required for the model	362887460	479522060	485454890	362945029	362141400	374560920	*

Note: The figure's overall low accuracy is due to the small CNN model used. Due to the large amount of fake data generated during global training, only the first 200 rounds were trained. To speed up the training, we set each federated training round to 1.

the generated features and the original data under the differential privacy mechanism.

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